



Holy Cross College

Semester 1 Examination 2019

Question/Answer Booklet

**Year 10 Science
(Adjusted)**

Please place your student identification label in this box

Student Name

SOLUTIONS

Student's Teacher

Time allowed for this paper

Reading time before commencing work: 5 minutes
Working time for paper: 60 minutes

Materials required/recommended for this paper

To be provided by the supervisor

This Question/Answer Booklet
Data Sheet

To be provided by the candidate

Standard items: pens, pencils, eraser, correction fluid, ruler, highlighters.

Special items: non-programmable calculators satisfying the conditions set by the School Curriculum and Standards Authority for this course.

Important note to candidates

No other items may be taken into the examination room. It is **your** responsibility to ensure that you do not have any unauthorised notes or other items of a non-personal nature in the examination room. If you have any unauthorised material with you, hand it to the supervisor **before** reading any further.

Structure of this paper

Section	Number of questions available	Number of questions to be answered	Suggested working time (minutes)	Marks available	Percentage of exam
Section One: Multiple Choice	20	20	15	20	26
Section Two: Short Answer	16	16	45	58	74
Section Three: Comprehension					
				78	100

Instructions to candidates

1. The rules for the conduct of examinations at Holy Cross College are detailed in the College Examination Policy. Sitting this examination implies that you agree to abide by these rules.
2. Write your answers in this Question/Answer Booklet.
3. Working or reasoning should be clearly shown when calculating or estimating answers.
4. You must be careful to confine your responses to the specific questions asked and to follow any instructions that are specific to a particular question.
5. Spare pages are included at the end of this booklet. They can be used for planning your responses and/or as additional space if required to continue an answer.
 - Planning: If you use the spare pages for planning, indicate this clearly at the top of the page.
 - Continuing an answer: If you need to use the space to continue an answer, indicate in the original answer space where the answer is continued, i.e. give the page number.
 - Fill in the number of the question(s) that you are continuing to answer at the top of the page.

SECTION 1 MULTIPLE CHOICE (20 marks)

Choose the best answer for each question in the Multiple Choice Question booklet.

Mark your selection with a cross.

- | | | | | |
|----|-----|-----|-----|-----|
| 1 | (a) | (b) | (c) | (d) |
| 2 | (a) | (b) | (c) | (d) |
| 3 | (a) | (b) | (c) | (d) |
| 4 | (a) | (b) | (c) | (d) |
| 5 | (a) | (b) | (c) | (d) |
| 6 | (a) | (b) | (c) | (d) |
| 7 | (a) | (b) | (c) | (d) |
| 8 | (a) | (b) | (c) | (d) |
| 9 | (a) | (b) | (c) | (d) |
| 10 | (a) | (b) | (c) | (d) |
| 11 | (a) | (b) | (c) | (d) |
| 12 | (a) | (b) | (c) | (d) |
| 13 | (a) | (b) | (c) | (d) |
| 14 | (a) | (b) | (c) | (d) |
| 15 | (a) | (b) | (c) | (d) |
| 16 | (a) | (b) | (c) | (d) |
| 17 | (a) | (b) | (c) | (d) |
| 18 | (a) | (b) | (c) | (d) |
| 19 | (a) | (b) | (c) | (d) |
| 20 | (a) | (b) | (c) | (d) |

Must have both.

SECTION 2

SHORT ANSWER (58 marks)

Attempt all questions. Write your answers in the spaces provided.

1. Use the words listed below to complete the following sentences. (7 marks)

protostar	billion	nebulae	giants	fusion
space	expansion			

Clouds of dust and gas in interstellar space are called nebulae.

According to the Big Bang Theory, the universe is about 13.7 billion years old.

The first stage of the Big Bang involved the formation of time and space.

Space then underwent rapid expansion.

Stars are formed when gravity causes gas and dust to come together and heat up.

Eventually a protostar is formed and then finally a star.

Stars like our sun evolve into red giants before they explode.

Stars generate their energy by nuclear fusion in which hydrogen changes to helium.

[1 mark each]

2. What does the Big Bang Theory state? (2 marks)

- In the beginning was a singularity. (1)
- This suddenly expands into space and time. (1)

3. Arrange the following stages in the life cycle of a star in the correct order from **earliest to latest**.

red giant, white dwarf, main sequence star, black dwarf

(2 marks)

main sequence star red giant white dwarf black dwarf
[$\frac{1}{2}$ mark each]

4. A racing car makes a 'yeee ... owww' sound as it speeds towards and then past an observer.

Choose the correct terms in the following statement, relating the sound to the blue and red shift of the Doppler effect and to the movement of galaxies towards and away from us. (**Circle the correct answer.**) (3 marks)

The 'yeee' is a higher / lower frequency, which corresponds to a

blue shift / red shift in light, which indicates movement towards / away

from us.

[1 mark each]

5. Complete the following table. Include the correct units. (Hint: Use sp, d, t triangle.) (3 marks)

Distance travelled	Time taken	Average speed
200 m	25 s	8 m/s
375 m	25 s	15 m/s
2750 m	132 s	20.8 m/s

[1 mark each]

6. How long will it take to complete a trip by car from Perth to Kalgoorlie, a distance of 595 km, travelling at an average speed of 85 km/h? (Give your answer in hours.) (Hint: Use sp, d, t triangle.) (3 marks)

$$sp = 85 \text{ km/h}$$

$$d = 595 \text{ km}$$

$$t = ?$$

$$t = \frac{d}{sp} \quad (1)$$

$$= \frac{595}{85} \quad (1)$$

$$= 7.0 \text{ hrs} \quad (1)$$

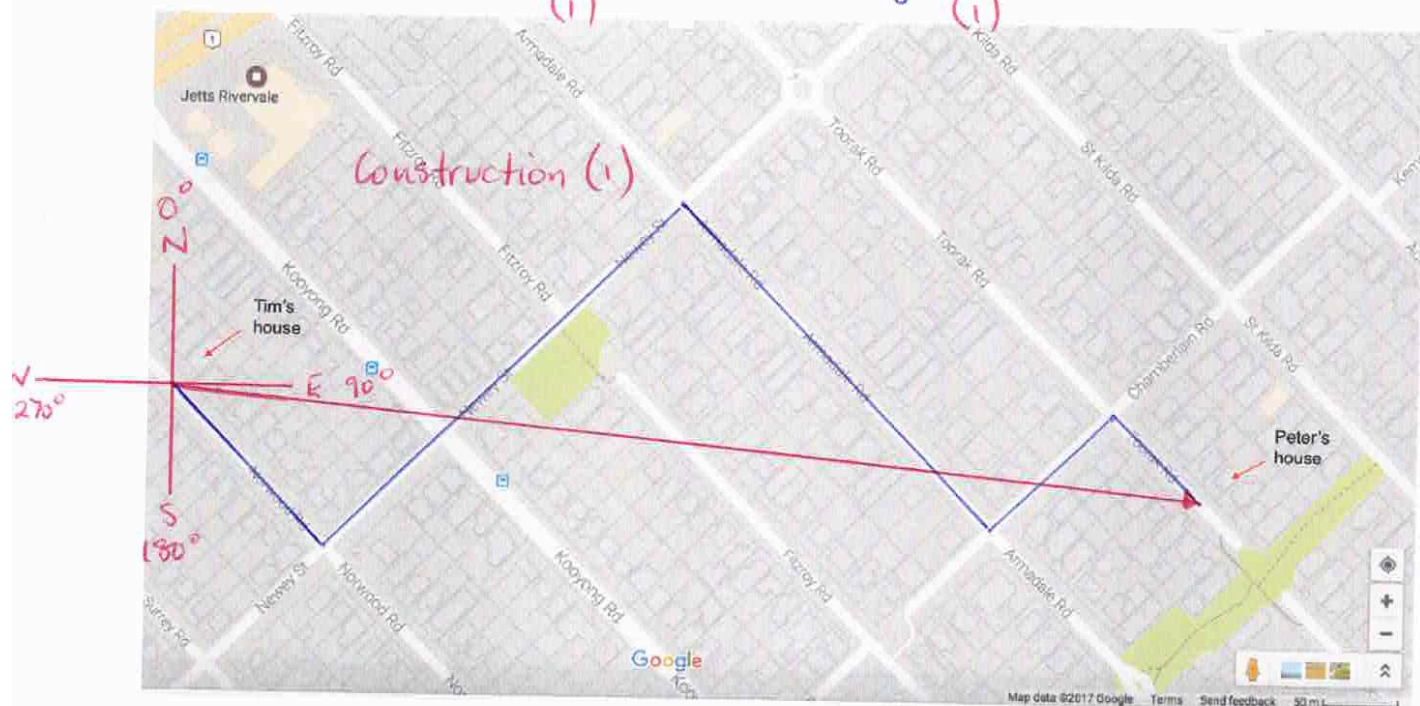
7. Tim and Peter are good friends who live in the Rivervale area. Tim was planning to walk from his place to Peter's and the map below shows their respective houses. Use the map to do the following.

- (a) **Trace on the map** the way you would walk to Peter's house. Using the scale, calculate (approximately) the **distance** you would walk. (2 marks)

965 m (± 30 m) (2)

- (b) Use the scale to determine the **displacement** and **bearing** from Tim's house to Peter's house. Remember to draw the compass directions on the map. (3 marks)

680 m (± 30 m) bearing $95^\circ (\pm 5^\circ)$
(1) (1)



8. Use the periodic table supplied to classify each of the elements below as either **metals**, **non-metals** or **metalloids**. (5 marks)

Mg, C, As, Kr, Fe, N, H, Al, Cu, B

Metal	Non-metal	Metalloid
Mg	C	As
Fe	Kr	B
Al	N	
Cu	H	

[$\frac{1}{2}$ mark each]

9. Complete the following table showing the number of particles of each element. (7 marks)

Element Name	Element symbol	Atomic number	Mass number	Number of Protons	Number of Neutrons	Number of Electrons
Carbon	C	6	12	6	6	6
Calcium	Ca	20	40	20	20	20
Hydrogen	H	1	2	1	1	1

10. Aluminium is used to make soft drink cans.

- (a) Is aluminium a **pure substance** or a **mixture**? (1 mark)

• pure substance (1)

- (b) Circle the type of bonding in aluminium. (1 mark)

metallic

covalent

ionic

(1)



- (c) One property that makes aluminium suitable to make soft drink cans is its malleability.

(i) Define malleability.

(1 mark)

• Can be hammered or rolled into sheets. (1)

(ii) Explain why it is important for aluminium to be malleable if it is to be used in the production of cans.

(1 mark)

• Need to be able to form the shape easily. (1)

(iii) Describe **one disadvantage** of aluminium being malleable.

(1 mark)

• Very soft and easily damaged or deformed. (1)

11. Describe **three** features of DNA structure (shape, what it is made of, etc.) (3 marks)

- Double-helix shape. (spiral) (1)
- Has bases bonded together. (1)
- Made up of nucleotides containing a phosphate group, a sugar group and a nitrogen base. (1)

12. Complete the complementary DNA strand for the sequence below.

(1 mark)

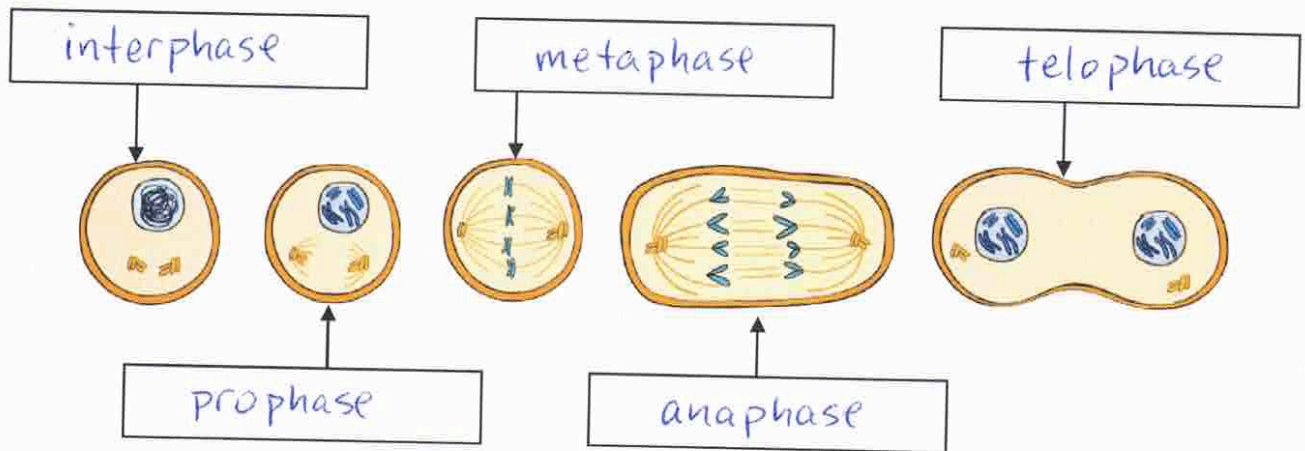
T A C G G A C T T G C A

A T G C C T G A A C G T (1)

13. Label the following stages of the cell cycle and mitosis using the words listed.

(5 marks)

metaphase, telophase, anaphase, interphase, prophase



[1 mark each]

14. Describe the role of the spindle fibres in mitosis.

(1 mark)

- To pull the chromosomes apart. (1)

15. Brown eyes are dominant to blue eyes.

- (a) A blue-eyed woman marries a man with brown eyes. Their first child has blue eyes. What does this tell us about the father's genotype? (1 mark)

- Bb - carries the b gene for blue. (1)

- (b) What chance does their second child have of inheriting brown eyes? Use a Punnett square to show your working. (2 marks)

♀ \ ♂	B	b
b	Bb	bb
b	Bb	bb

(1)

Genotype: Bb Bb bb bb

Phenotype: 50% brown 50% blue

∴ 50% chance (1)

16. (a) A red flower (RR) and a white flower (WW) are crossed to produce a pink flower. What is the name of this type of inheritance? (1 mark)

• Incomplete dominance

- (b) Two pink flowers are mixed. Complete a Punnett square to show the possible offspring. (2 marks)

♀ \ ♂	R	W
R	RR	RW
W	RW	WW

Genotype: RR RW RW WW
 Phenotype: red pink white (1)

END OF EXAMINATION

ADDITIONAL WORKING SPACE